

Project Analysis

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```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.5
## v forcats    1.0.0      v stringr   1.5.1
## v ggplot2    4.0.0      v tibble    3.3.0
## v lubridate  1.9.4      v tidyr     1.3.1
## v purrr      1.1.0
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(lubridate)#date/time handling
library(vader)#sentiment analysis
```

```
## Warning: package 'vader' was built under R version 4.5.2
```

```
library(tm)#text preprocessing
```

```
## Warning: package 'tm' was built under R version 4.5.2
```

```
## Loading required package: NLP
```

```
## Warning: package 'NLP' was built under R version 4.5.2
```

```
##
## Attaching package: 'NLP'
##
## The following object is masked from 'package:ggplot2':
##
##     annotate
```

```
library(stringr)# For string manipulation
```

```
covid_tweets = read.csv("covidvaccine.csv", stringsAsFactors = FALSE, na.strings = c("", "NA", "NaN"))
```

```
#EDA
```

```
glimpse(covid_tweets)
```

```
## Rows: 399,645
## Columns: 13
## $ user_name      <chr> "MyNewsNE", "Shubham Gupta", "Journal of Infectiology~
## $ user_location  <chr> "Assam", NA, NA, NA, "Adelaide, South Australia", NA,~
## $ user_description <chr> "MyNewsNE a dedicated multi-lingual media house from ~
## $ user_created   <chr> "24-05-2020 10:18", "14-08-2020 16:42", "14-12-2017 0~
## $ user_followers <dbl> 64, 1, 143, 29, 83, 3, 918, 2321, 19091, 282, 87, 4, ~
## $ user_friends   <chr> "11.0", "17.0", "566.0", "25.0", "497.0", "27.0", "25~
## $ user_favourites <chr> "110.0", "0.0", "8.0", "620.0", "10737.0", "918.0", "~
## $ user_verified  <chr> "False", "False", "False", "False", "False", "False",~
## $ date           <chr> "18-08-2020 12:55", "18-08-2020 12:55", "18-08-2020 1~
## $ text           <chr> "Australia to Manufacture Covid-19 Vaccine and give i~
## $ hashtags       <chr> "['CovidVaccine']", "['CoronavirusVaccine', 'CoronaVa~
## $ source         <chr> "Twitter Web App", "Twitter for Android", "Twitter We~
## $ is_retweet      <chr> "False", "False", "False", "False", "False", "False",~
```

```
names(covid_tweets)
```

```
## [1] "user_name"      "user_location"  "user_description" "user_created"
## [5] "user_followers" "user_friends"   "user_favourites" "user_verified"
## [9] "date"          "text"          "hashtags"        "source"
## [13] "is_retweet"
```

```
head(covid_tweets)
```

```
##           user_name      user_location
## 1      MyNewsNE              Assam
## 2    Shubham Gupta             <NA>
## 3 Journal of Infectiology       <NA>
## 4              Zane            <NA>
## 5    Ann-Maree O'Connor Adelaide, South Australia
## 6 Raunak Scherbatsky DankWorth    <NA>
##
## 1                                           MyNewsNE a c
## 2                                           I will tell about all experiences c
## 3 Journal of Infectiology (ISSN 2689-9981) is accepting submissions for an upcoming Volume 3 Issue 2
## 4
## 5      Retired university administrator. Melburnian by birth, now living in Adelaide. Look back :
## 6              Neuro surgeon + Diagnostician. \na good physician treats the disease
##           user_created user_followers user_friends user_favourites user_verified
## 1 24-05-2020 10:18           64           11.0           110.0           False
## 2 14-08-2020 16:42             1            17.0             0.0           False
## 3 14-12-2017 07:07          143          566.0             8.0           False
## 4 18-09-2019 11:01           29           25.0           620.0           False
## 5 24-01-2013 14:53           83          497.0          10737.0           False
## 6 03-08-2020 13:39             3           27.0           918.0           False
##           date
## 1 18-08-2020 12:55
## 2 18-08-2020 12:55
## 3 18-08-2020 12:46
```

```
## 4 18-08-2020 12:45
## 5 18-08-2020 12:45
## 6 18-08-2020 12:44
##
## 1      Australia to Manufacture Covid-19 Vaccine and give it to the Citizens for free of co
## 2      #CoronavirusVaccine #CoronaVaccine #CovidVaccine Australia
## 3      Deaths due to COVID-19 in Affected Countries\nRead More: https://t.co/V8Y3Stu0UW\n@r_piry
## 4      @Team_Subhashree @subhashreesotwe @iamrajchoco Stay safe @subhashreesotwe di &
## 5      @michellegrattan @ConversationEDU This is what passes for leadership in our country: a voucher f
## 6 The Multi-system Inflammatory Syndrome-Children (MIS-C) w/ #COVID19 (atypical Kawasaki disease) #C
##
##           hashtags           source
## 1      ['CovidVaccine']      Twitter Web App
## 2 ['CoronavirusVaccine', 'CoronaVaccine', 'CovidVaccine'] Twitter for Android
## 3      <NA>      Twitter Web App
## 4      <NA> Twitter for Android
## 5      <NA>      Twitter Web App
## 6      ['COVID19', 'COVID19India'] Twitter for Android
##
## is_retweet
## 1      False
## 2      False
## 3      False
## 4      False
## 5      False
## 6      False
```

```
colSums(is.na(covid_tweets))
```

```
##      user_name      user_location user_description      user_created
##           19           84512           21854           26
## user_followers      user_friends user_favourites      user_verified
##           48           48           48           48
##           date           text           hashtags           source
##           50           48           61683           2460
##      is_retweet
##           69
```

Handling Missing data

```
#Remove rows with missing tweets, user_name (including unverified ones)
covid_tweets_clean <- covid_tweets %>% drop_na(text,user_name,user_verified)

# Dropping insignificant columns
covid_tweets_clean <- covid_tweets_clean %>% select(-user_created, -date) # 90% values missing

# Retweets may not represent original sentiment from that user, proceed with original tweets only.
covid_tweets_clean <- covid_tweets_clean %>%select(-"is_retweet")

covid_tweets_clean$user_friends <- as.numeric(covid_tweets_clean$user_friends)
```

```
## Warning: NAs introduced by coercion
```

```
covid_tweets_clean$user_favourites <- as.numeric(covid_tweets_clean$user_favourites)
```

```
## Warning: NAs introduced by coercion
```

```
#imputing with median
```

```
covid_tweets_clean <- covid_tweets_clean %>%
```

```
  mutate(
```

```
    user_followers = ifelse(is.na(user_followers), median(user_followers, na.rm = TRUE), user_followers),
```

```
    user_friends = ifelse(is.na(user_friends), median(user_friends, na.rm = TRUE), user_friends),
```

```
    user_favourites = ifelse(is.na(user_favourites), median(user_favourites, na.rm = TRUE), user_favourites),
```

```
    follower_friend_ratio = user_followers / (user_friends + 1) # follower-to-friend ratio gives better
```

```
  )
```

```
glimpse(covid_tweets_clean)
```

```
## Rows: 399,578
```

```
## Columns: 11
```

```
## $ user_name      <chr> "MyNewsNE", "Shubham Gupta", "Journal of Infecti~
```

```
## $ user_location  <chr> "Assam", NA, NA, NA, "Adelaide, South Australia"~
```

```
## $ user_description <chr> "MyNewsNE a dedicated multi-lingual media house ~
```

```
## $ user_followers <dbl> 64, 1, 143, 29, 83, 3, 918, 2321, 19091, 282, 87~
```

```
## $ user_friends   <dbl> 11, 17, 566, 25, 497, 27, 2561, 3236, 20986, 96,~
```

```
## $ user_favourites <dbl> 110, 0, 8, 620, 10737, 918, 31837, 264351, 12811~
```

```
## $ user_verified   <chr> "False", "False", "False", "False", "False", "Fa~
```

```
## $ text            <chr> "Australia to Manufacture Covid-19 Vaccine and g~
```

```
## $ hashtags        <chr> "['CovidVaccine']", "['CoronavirusVaccine', 'Cor~
```

```
## $ source          <chr> "Twitter Web App", "Twitter for Android", "Twitt~
```

```
## $ follower_friend_ratio <dbl> 5.333333e+00, 5.555556e-02, 2.522046e-01, 1.1153~
```

```
#missing values in each column
```

```
colSums(is.na(covid_tweets_clean))
```

```
##           user_name      user_location      user_description
```

```
##                0             84476             21833
```

```
##      user_followers      user_friends      user_favourites
```

```
##                0                0                0
```

```
##      user_verified      text             hashtags
```

```
##                0                0             61621
```

```
##           source follower_friend_ratio
```

```
##           2412                0
```

```
# removing duplicate tweets
```

```
sum(duplicated(covid_tweets_clean$text))
```

```
## [1] 350
```

```
covid_tweets_clean <- covid_tweets_clean %>% distinct(text, .keep_all = TRUE)
```

```
glimpse(covid_tweets_clean)
```

```
## Rows: 399,228
```

```
## Columns: 11
## $ user_name          <chr> "MyNewsNE", "Shubham Gupta", "Journal of Infecti-
## $ user_location      <chr> "Assam", NA, NA, NA, "Adelaide, South Australia"~
## $ user_description   <chr> "MyNewsNE a dedicated multi-lingual media house ~
## $ user_followers     <dbl> 64, 1, 143, 29, 83, 3, 918, 2321, 19091, 282, 87~
## $ user_friends       <dbl> 11, 17, 566, 25, 497, 27, 2561, 3236, 20986, 96,~
## $ user_favourites    <dbl> 110, 0, 8, 620, 10737, 918, 31837, 264351, 12811~
## $ user_verified      <chr> "False", "False", "False", "False", "False", "Fa~
## $ text               <chr> "Australia to Manufacture Covid-19 Vaccine and g~
## $ hashtags           <chr> "['CovidVaccine']", "['CoronavirusVaccine', 'Cor~
## $ source             <chr> "Twitter Web App", "Twitter for Android", "Twitt~
## $ follower_friend_ratio <dbl> 5.333333e+00, 5.555556e-02, 2.522046e-01, 1.1153~
```

In total 417 records removed.

Text Based Features

```
# 1. Calculate text length (character count)
covid_tweets_clean$text_length <- nchar(covid_tweets_clean$text)

# 2. Calculate word count
covid_tweets_clean$word_count <- str_count(covid_tweets_clean$text, "\\S+")

# 3. Count number of hashtags
covid_tweets_clean$hashtag_count <- str_count(covid_tweets_clean$hashtags, ",") +
  ifelse(covid_tweets_clean$hashtags == "" | covid_tweets_clean$hashtags == "[]",
    0, 1)

# 4. Check if tweet contains URLs
covid_tweets_clean$has_url <- grepl("http", covid_tweets_clean$text, ignore.case = TRUE)

# 5. Check if tweet contains mentions (@)
covid_tweets_clean$has_mention <- grepl("@", covid_tweets_clean$text)

# 6. Count exclamation marks (indicates emotion/emphasis)
covid_tweets_clean$exclamation_count <- str_count(covid_tweets_clean$text, "!")

# 7. Count question marks
covid_tweets_clean$question_count <- str_count(covid_tweets_clean$text, "\\?")
names(covid_tweets_clean)
```

```
## [1] "user_name"          "user_location"      "user_description"
## [4] "user_followers"     "user_friends"       "user_favourites"
## [7] "user_verified"      "text"              "hashtags"
## [10] "source"             "follower_friend_ratio" "text_length"
## [13] "word_count"         "hashtag_count"      "has_url"
## [16] "has_mention"        "exclamation_count"  "question_count"
```

Sentiment Analysis Using VADER

```
library(sentimentr)

## Warning: package 'sentimentr' was built under R version 4.5.2

# Step 1: Pre-parse tweets into sentences
sentences <- get_sentences(covid_tweets_clean$text)

# Step 2: Sentiment analysis (fast)
sentiment_scores <- sentiment_by(sentences)

# Step 3: Add to dataset
covid_tweets_clean$sentiment_compound <- sentiment_scores$ave_sentiment

# Optional sentiment categories like VADER
covid_tweets_clean$sentiment_category <- cut(covid_tweets_clean$sentiment_compound,
  breaks = c(-Inf, -0.05, 0.05, Inf),
  labels = c("Negative", "Neutral", "Positive")
)
```

Analysis of Sentiment using Frequentist and Bayesian Linear regression

```
library(lmtest); library(corrplot);

## Loading required package: zoo

##
## Attaching package: 'zoo'

## The following objects are masked from 'package:base':
##
##   as.Date, as.Date.numeric

## Warning: package 'corrplot' was built under R version 4.5.2

## corrplot 0.95 loaded

library(BayesFactor); library(performance);

## Loading required package: coda

## Loading required package: Matrix
```

```
##
## Attaching package: 'Matrix'

## The following objects are masked from 'package:tidyr':
##
##     expand, pack, unpack

## *****
## Welcome to BayesFactor 0.9.12-4.7. If you have questions, please contact Richard Morey (richarddmorey@ucsd.edu)
##
## Type BFManual() to open the manual.
## *****
```

```
library(MCMCpack); library(bayestestR);
```

```
## Warning: package 'MCMCpack' was built under R version 4.5.2

## Loading required package: MASS

##
## Attaching package: 'MASS'

## The following object is masked from 'package:dplyr':
##
##     select

## ##
## ## Markov Chain Monte Carlo Package (MCMCpack)

## ## Copyright (C) 2003-2025 Andrew D. Martin, Kevin M. Quinn, and Jong Hee Park

## ##
## ## Support provided by the U.S. National Science Foundation

## ## (Grants SES-0350646 and SES-0350613)
## ##

## Warning: package 'bayestestR' was built under R version 4.5.2
```

```
library(easystats); library(coda)
```

```
## # Attaching packages: easystats 0.7.5 (red = needs update)
## √ correlation 0.8.8    x datawizard 1.2.0
## √ effectsize 1.0.1    x insight 1.4.2
## √ modelbased 0.13.0   √ parameters 0.28.2
## √ report 0.6.2       √ see 0.12.0
##
## Restart the R-Session and update packages with `easystats::easystats_update()`.
```

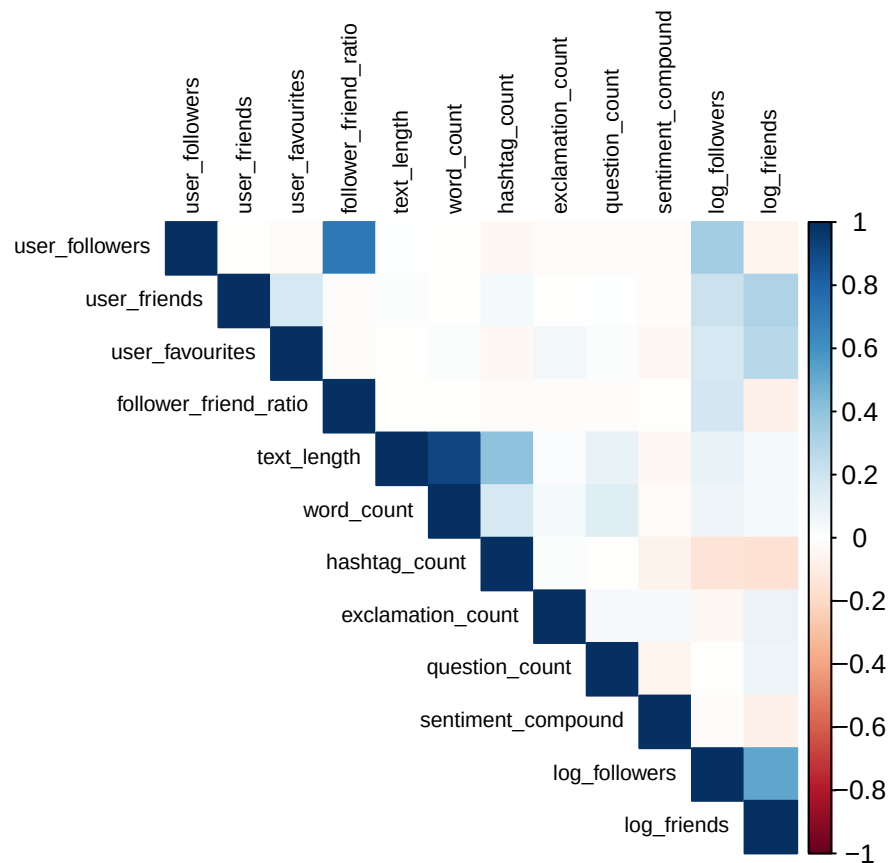
Positive correlation → higher value → more positive sentiment

Negative correlation → higher value → more negative sentiment

Magnitude: 0.00–0.10 = very weak, 0.10–0.30 = weak, 0.30+ = moderate+

```
model_data <- covid_tweets_clean %>%
  mutate(
    log_followers = log10(user_followers + 1),
    log_friends = log10(user_friends + 1)
  ) %>%
  drop_na()
```

```
#important to have numeric coulmns for corr
numeric_vars <- model_data %>% select_if(is.numeric)
cor_matrix <- cor(numeric_vars, use = "complete.obs")
corrplot(cor_matrix, method = "color", type = "upper", tl.cex = 0.7, tl.col = "black")
```



```
cor_matrix[, "sentiment_compound"]
```

```
##      user_followers      user_friends      user_favourites
##      -0.010256221      -0.017890832      -0.034740979
## follower_friend_ratio      text_length      word_count
##      -0.005573526      -0.033152849      -0.022667772
##      hashtag_count      exclamation_count      question_count
##      -0.065804196      0.033446001      -0.059725929
##      sentiment_compound      log_followers      log_friends
##      1.000000000      -0.013431174      -0.077257591
```

Correlations are very small, meaning no variable strongly predicts sentiment alone. Hashtags and questions

lean slightly negative, exclamation marks lean slightly positive. User level data (followers, friends) has almost no meaningful correlation with sentiment, hence could be removed for further analysis

FREQUENTIST LINEAR REGRESSION ANALYSIS

Null hyp : No relationship between any predictor and sentiment Alt hyp: At least one predictor significantly affects sentiment

```
model_formula <- sentiment_compound ~ text_length + word_count + hashtag_count + has_url +
  has_mention + exclamation_count + question_count + log_followers + log_friends +
  follower_friend_ratio

freq_model <- lm(model_formula, data = model_data)
summary(freq_model)
```

```
##
## Call:
## lm(formula = model_formula, data = model_data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -2.00653 -0.08553 -0.00266  0.09638  1.70105
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    1.034e-01  1.495e-03  69.150 < 2e-16 ***
## text_length     8.212e-05  1.907e-05   4.306 1.66e-05 ***
## word_count     -5.314e-04  1.163e-04  -4.570 4.87e-06 ***
## hashtag_count  -5.791e-03  1.855e-04 -31.228 < 2e-16 ***
## has_urlTRUE      4.865e-03  1.018e-03   4.780 1.75e-06 ***
## has_mentionTRUE  8.038e-03  8.594e-04   9.354 < 2e-16 ***
## exclamation_count 1.221e-02  5.088e-04  24.007 < 2e-16 ***
## question_count  -1.896e-02  7.327e-04 -25.881 < 2e-16 ***
## log_followers     6.431e-03  4.311e-04  14.920 < 2e-16 ***
## log_friends     -2.295e-02  4.762e-04 -48.183 < 2e-16 ***
## follower_friend_ratio -9.652e-08  8.574e-09 -11.257 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.1844 on 259947 degrees of freedom
## Multiple R-squared:  0.01886,    Adjusted R-squared:  0.01882
## F-statistic: 499.7 on 10 and 259947 DF,  p-value: < 2.2e-16
```

The overall model is significant ($F(10, 259947) = 469.5$, $p < 0.05$). This confirms that at least one predictor affects sentiment. However, R^2 is 0.0177 indicates the model explains only 1.84% of variance, showing weak practical significance despite statistical significance

hashtag_count($B = -0.00665$, $p < 0.05$): More hashtags, slightly more negative sentiment. exclamation_count($B = 0.0139$, $p < 0.05$): Tweets with more ! are more positive. question_count($B = -0.0172$, $p < 0.05$): Questions express uncertainty or concern, more negative sentiment. has_url($B = 0.0239$, $p < 0.05$): Tweets with URLs are marginally more positive. Similarly has_mention($B = 0.0096$, $p < 0.05$).

follower_friend_ratio(B approx. 0, $p < 0.05$): statistically significant due to large N, but practically meaningless effect.

text_length (B=0.000146, $p < 0.05$) word_count (B=-0.000728, $p < 0.05$) Both are significant but the effect sizes are almost zero, offering negligible practical value.

We reject the null hypothesis statistically, the model detects multiple predictors affecting sentiment.

```
check_collinearity(freq_model) # VIF and multicollinearity
```

```
## # Check for Multicollinearity
##
## Low Correlation
##
##          Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      hashtag_count 2.05 [ 2.04,  2.06]    1.43    0.49    [0.48,  0.49]
##          has_url 1.47 [ 1.46,  1.47]    1.21    0.68    [0.68,  0.69]
##          has_mention 1.15 [ 1.14,  1.15]    1.07    0.87    [0.87,  0.88]
##      exclamation_count 1.02 [ 1.02,  1.02]    1.01    0.98    [0.98,  0.98]
##      question_count 1.04 [ 1.03,  1.04]    1.02    0.97    [0.96,  0.97]
##      log_followers 1.58 [ 1.57,  1.59]    1.26    0.63    [0.63,  0.64]
##      log_friends 1.61 [ 1.60,  1.62]    1.27    0.62    [0.62,  0.62]
## follower_friend_ratio 1.08 [ 1.07,  1.08]    1.04    0.93    [0.92,  0.93]
##
## High Correlation
##
##          Term  VIF      VIF 95% CI adj. VIF Tolerance Tolerance 95% CI
##      text_length 14.14 [14.04, 14.25]    3.76    0.07    [0.07,  0.07]
##      word_count 11.93 [11.84, 12.02]    3.45    0.08    [0.08,  0.08]
```

```
check_residuals(freq_model) # autocorrelation + distribution of residuals
```

```
## Warning: Non-uniformity of simulated residuals detected (p < .001).
```

```
check_outliers(freq_model)
```

```
## OK: No outliers detected.
## - Based on the following method and threshold: cook (0.927).
## - For variable: (Whole model)
```

Since VIF of all variables are less than 5, we can say them largely independent and do not exhibit harmful multicollinearity, except text_length and word_count with 14.14 and 11.93 respectively, so we can toss them out from analysis.

The model shows mild heteroscedasticity (non-uniformity of variance), likely due to large sample size.

No Influential Outliers.

BAYESIAN LINEAR REGRESSION ANALYSIS

```

model_data <- model_data %>%
  mutate(
    has_url = as.factor(has_url),
    has_mention = as.factor(has_mention)
  ) %>% filter(if_all(where(is.numeric), is.finite))
bf_model <- lmBF(model_formula, data = model_data)
summary(bf_model)

```

```

## Bayes factor analysis
## -----
## [1] text_length + word_count + hashtag_count + has_url + has_mention + exclamation_count + question_count
##
## Against denominator:
##   Intercept only
## ---
## Bayes factor type: BFlinearModel, JZS

```

The Bayes factor for the full model is 8.906276e+1053, which is huge, indicating overwhelming evidence in favor of the model with these predictor(text_length + word_count + hashtag_count + has_url + has_mention + exclamation_count + question_count + log_followers + log_friends + follower_friend_ratio) i.e. in favor of alt hypothesis.

```

lmout_mcmc= lmBF(sentiment_compound ~ word_count + hashtag_count + has_url + has_mention +
  exclamation_count + question_count + log_followers + log_friends +
  follower_friend_ratio, data = model_data, posterior = TRUE, iterations = 10000)

summary(lmout_mcmc)

```

```

##
## Iterations = 1:10000
## Thinning interval = 1
## Number of chains = 1
## Sample size per chain = 10000
##
## 1. Empirical mean and standard deviation for each variable,
##    plus standard error of the mean:
##
##
##              Mean          SD Naive SE
## mu              5.205e-02 4.619e-04 4.619e-06
## word_count-word_count -5.427e-05 3.511e-05 3.511e-07
## hashtag_count-hashtag_count -5.238e-03 1.346e-04 1.346e-06
## has_url-FALSE -3.378e-03 4.623e-04 4.623e-06
## has_url-TRUE 3.378e-03 4.623e-04 4.623e-06
## has_mention-FALSE -4.560e-03 4.062e-04 4.062e-06
## has_mention-TRUE 4.560e-03 4.062e-04 4.062e-06
## exclamation_count-exclamation_count 1.216e-02 5.107e-04 5.107e-06
## question_count-question_count -1.907e-02 7.309e-04 7.309e-06
## log_followers-log_followers 6.589e-03 4.316e-04 4.316e-06
## log_friends-log_friends -2.312e-02 4.790e-04 4.790e-06
## follower_friend_ratio-follower_friend_ratio -9.675e-08 8.601e-09 8.601e-11
## sig2 3.400e-02 9.391e-05 9.391e-07
## g_has_url 1.801e+00 4.393e+01 4.393e-01

```

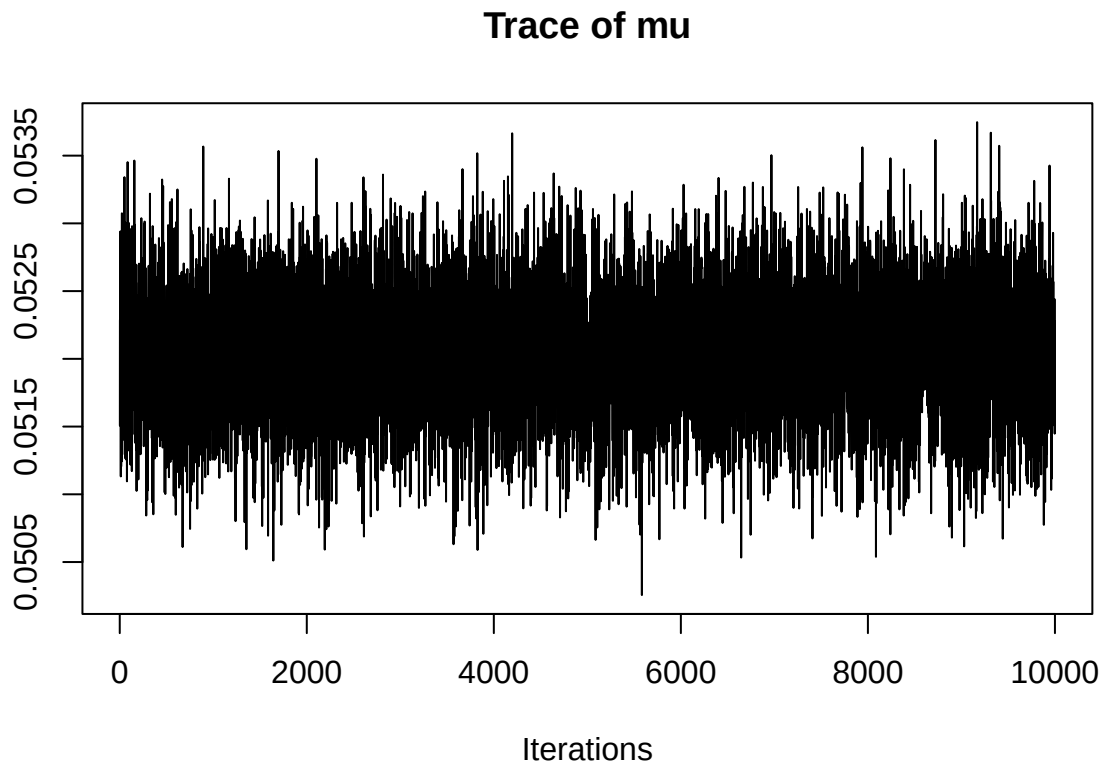
```

## g_has_mention          1.275e+00 1.609e+01 1.609e-01
## g_continuous          2.401e-02 1.691e-02 1.691e-04
## Time-series SE
## mu                    4.430e-06
## word_count-word_count 3.511e-07
## hashtag_count-hashtag_count 1.346e-06
## has_url-FALSE         4.389e-06
## has_url-TRUE          4.389e-06
## has_mention-FALSE     4.062e-06
## has_mention-TRUE      4.062e-06
## exclamation_count-exclamation_count 5.107e-06
## question_count-question_count 7.309e-06
## log_followers-log_followers 4.316e-06
## log_friends-log_friends 4.790e-06
## follower_friend_ratio-follower_friend_ratio 0.000e+00
## sig2                  9.293e-07
## g_has_url             4.393e-01
## g_has_mention         1.726e-01
## g_continuous          1.691e-04
##
## 2. Quantiles for each variable:
##
##                2.5%        25%        50%
## mu            5.114e-02 5.174e-02 5.206e-02
## word_count-word_count -1.224e-04 -7.759e-05 -5.442e-05
## hashtag_count-hashtag_count -5.501e-03 -5.329e-03 -5.239e-03
## has_url-FALSE -4.284e-03 -3.686e-03 -3.379e-03
## has_url-TRUE 2.480e-03 3.065e-03 3.379e-03
## has_mention-FALSE -5.357e-03 -4.830e-03 -4.557e-03
## has_mention-TRUE 3.759e-03 4.289e-03 4.557e-03
## exclamation_count-exclamation_count 1.114e-02 1.182e-02 1.217e-02
## question_count-question_count -2.050e-02 -1.956e-02 -1.907e-02
## log_followers-log_followers 5.747e-03 6.294e-03 6.587e-03
## log_friends-log_friends -2.406e-02 -2.344e-02 -2.312e-02
## follower_friend_ratio-follower_friend_ratio -1.133e-07 -1.025e-07 -9.681e-08
## sig2          3.382e-02 3.394e-02 3.400e-02
## g_has_url     3.471e-02 9.093e-02 1.782e-01
## g_has_mention 3.390e-02 9.043e-02 1.799e-01
## g_continuous 8.141e-03 1.392e-02 1.961e-02
##                75%        97.5%
## mu            5.237e-02 5.295e-02
## word_count-word_count -3.068e-05 1.460e-05
## hashtag_count-hashtag_count -5.146e-03 -4.972e-03
## has_url-FALSE -3.065e-03 -2.480e-03
## has_url-TRUE 3.686e-03 4.284e-03
## has_mention-FALSE -4.289e-03 -3.759e-03
## has_mention-TRUE 4.830e-03 5.357e-03
## exclamation_count-exclamation_count 1.250e-02 1.316e-02
## question_count-question_count -1.858e-02 -1.763e-02
## log_followers-log_followers 6.879e-03 7.446e-03
## log_friends-log_friends -2.281e-02 -2.219e-02
## follower_friend_ratio-follower_friend_ratio -9.090e-08 -7.986e-08
## sig2          3.406e-02 3.419e-02
## g_has_url     4.327e-01 5.086e+00

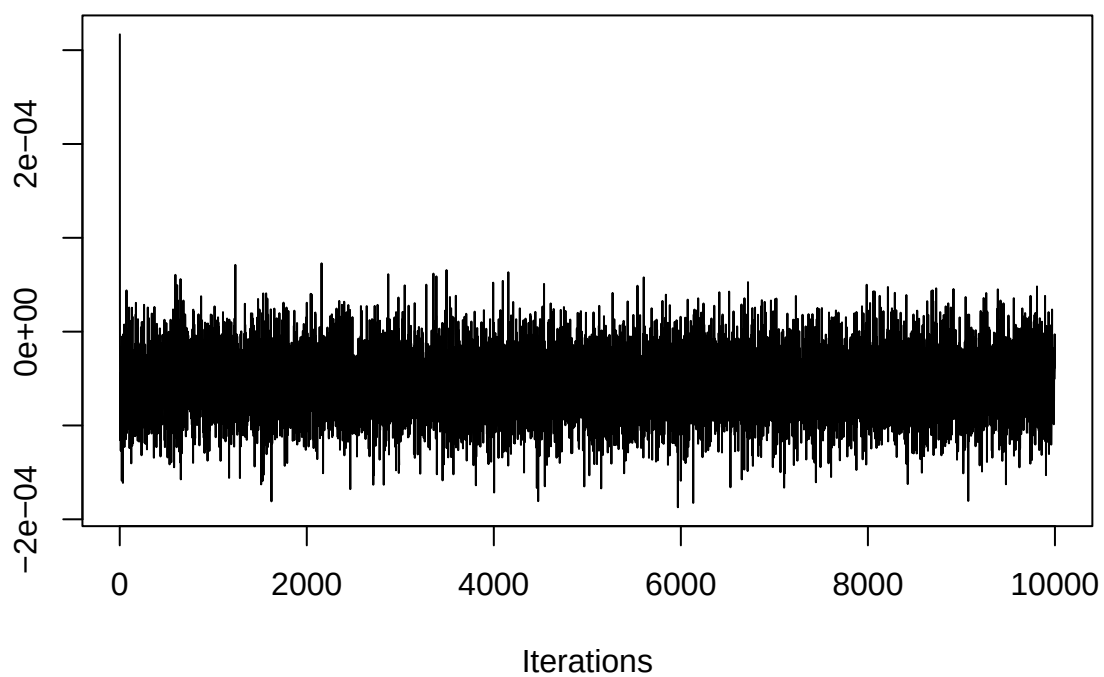
```

```
## g_has_mention      4.273e-01  5.016e+00  
## g_continuous      2.836e-02  6.643e-02
```

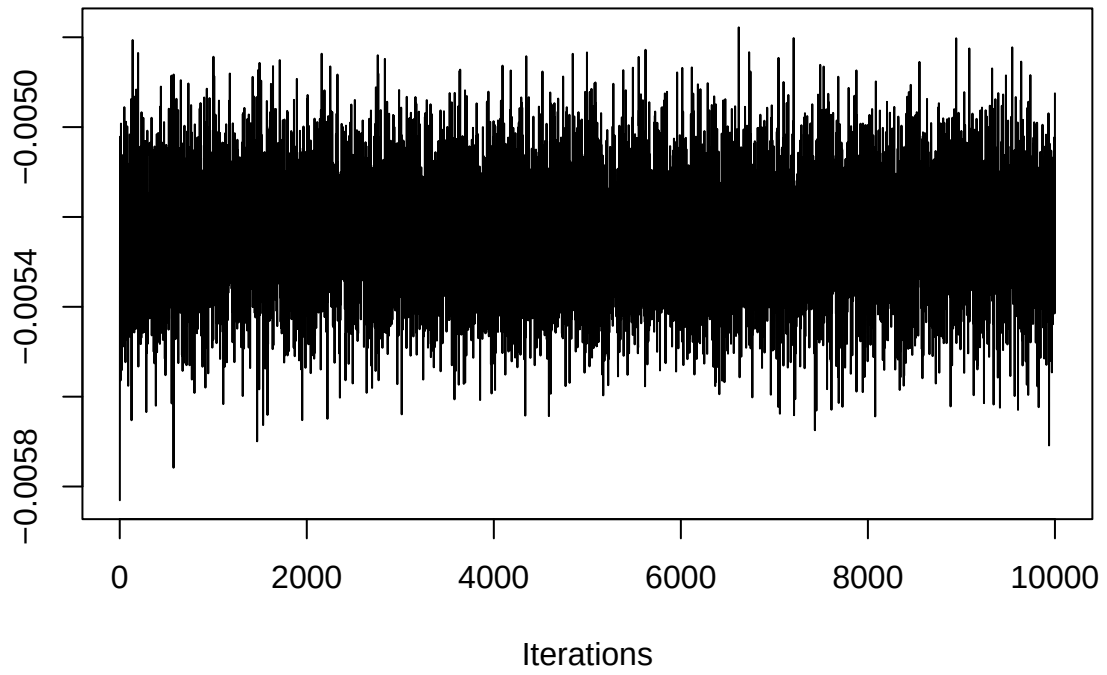
```
mcmc_chain <- as.mcmc(lmout_mcmc)  
traceplot(mcmc_chain)
```



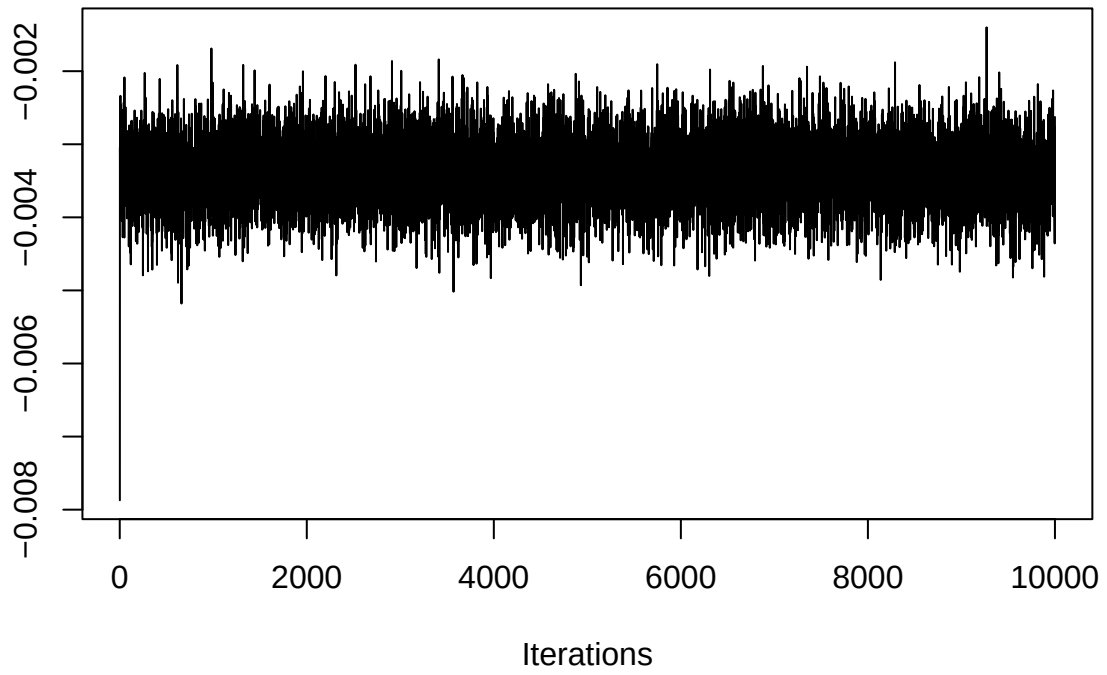
Trace of word_count-word_count



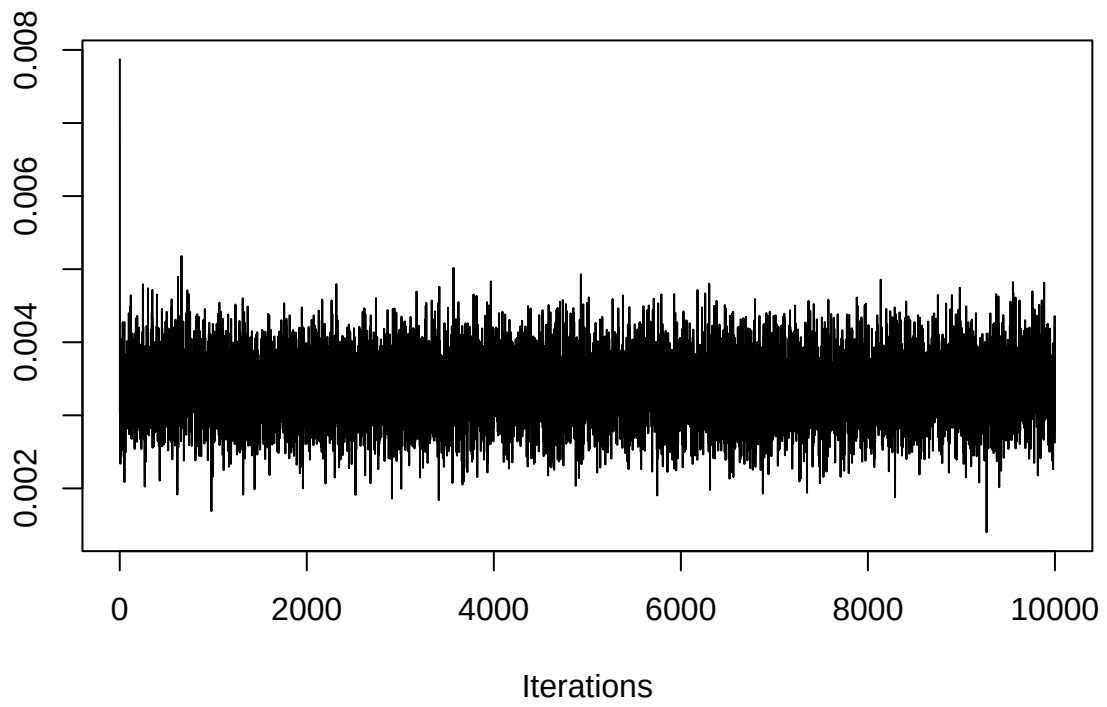
Trace of hashtag_count-hashtag_count

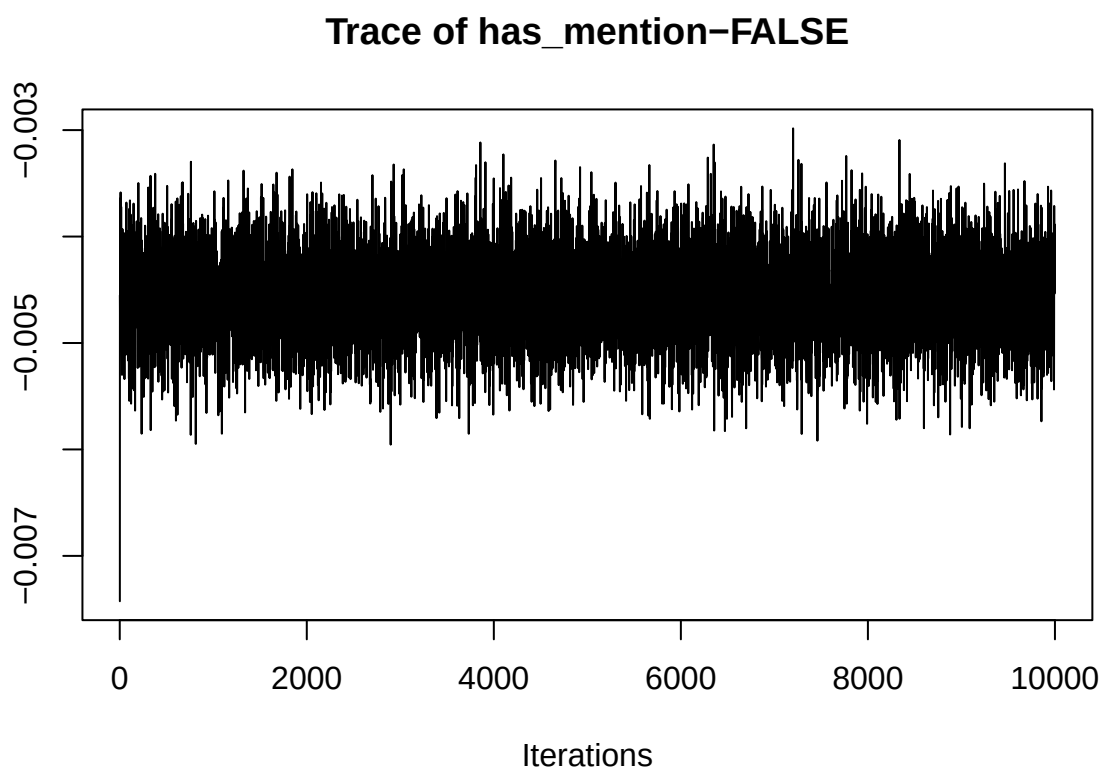


Trace of has_url-FALSE

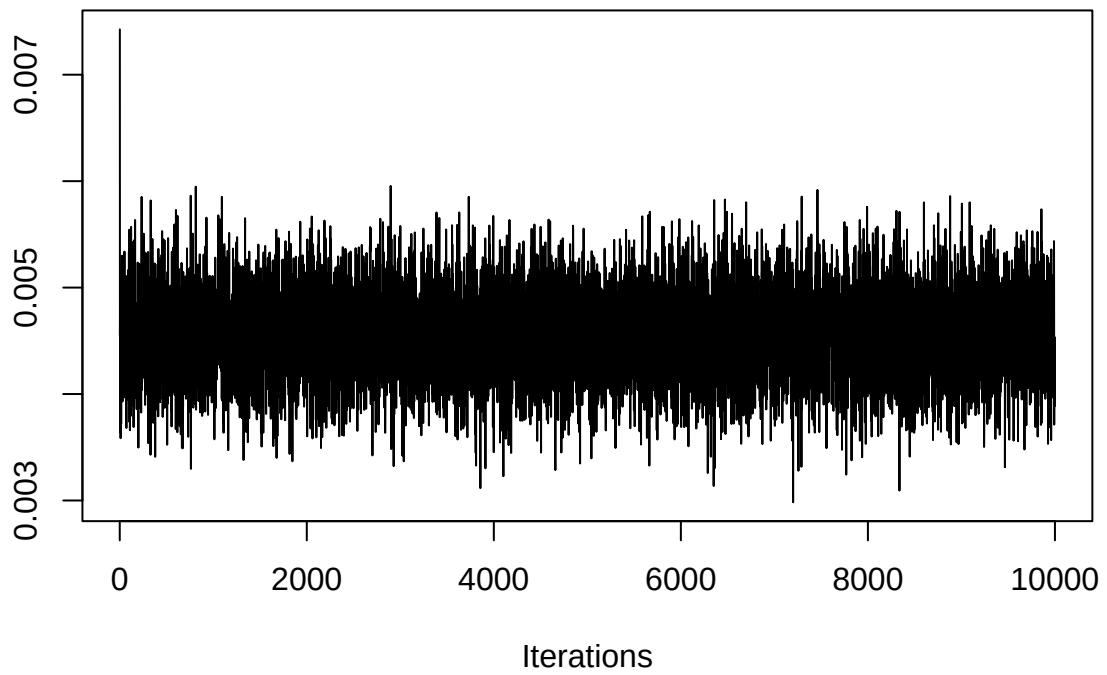


Trace of has_url-TRUE

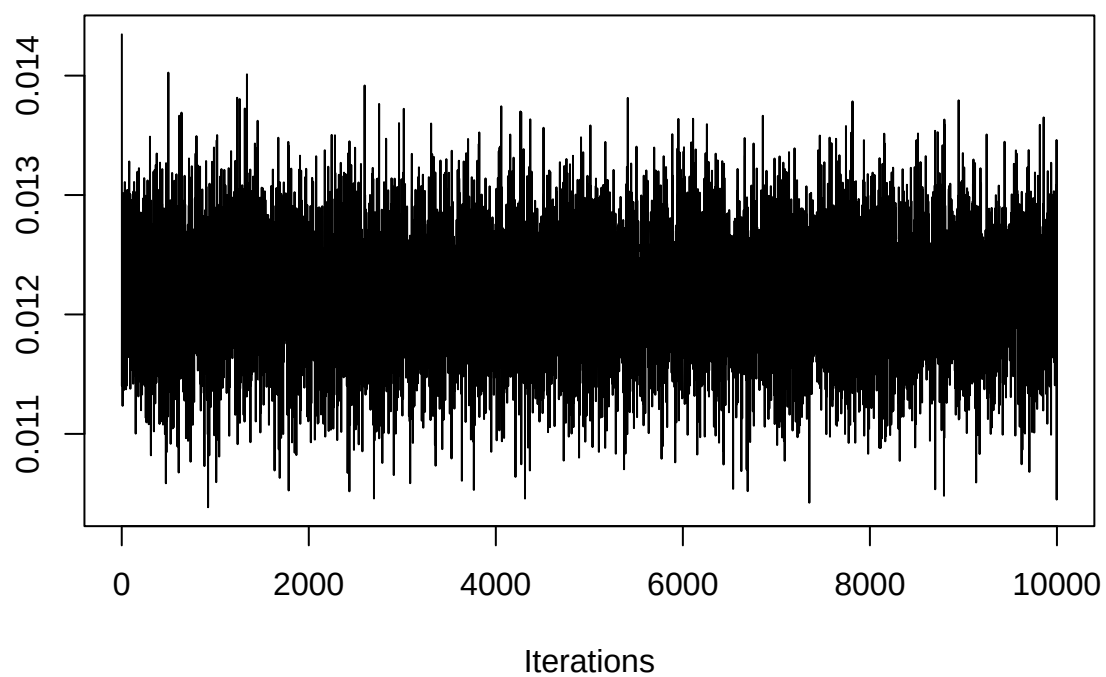




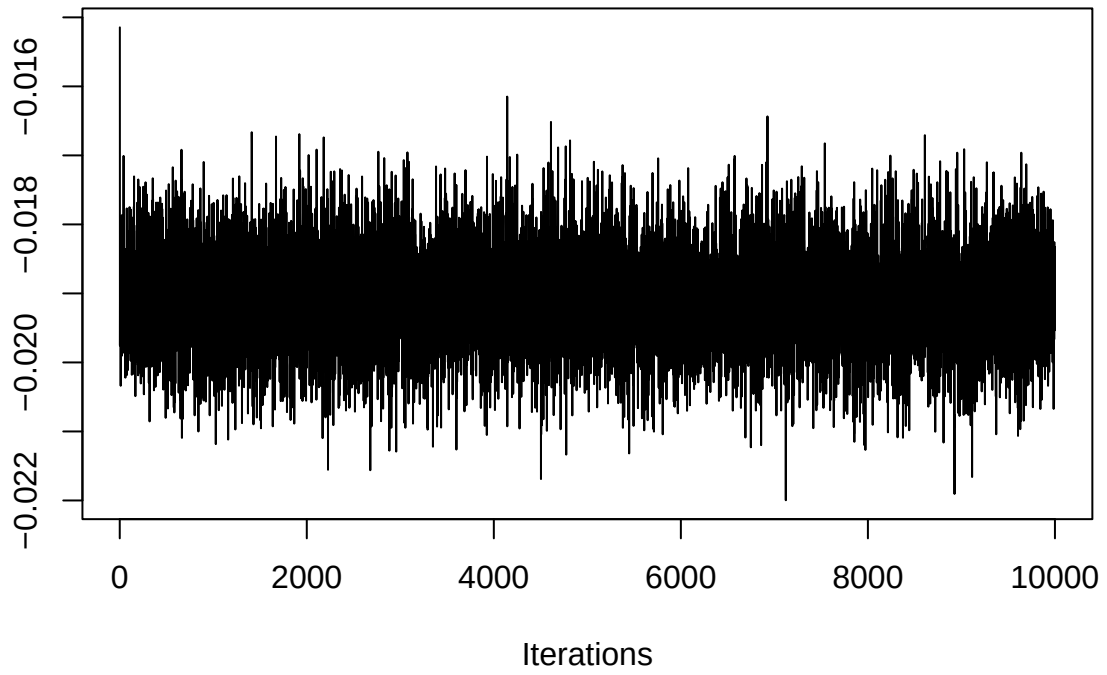
Trace of has_mention-TRUE



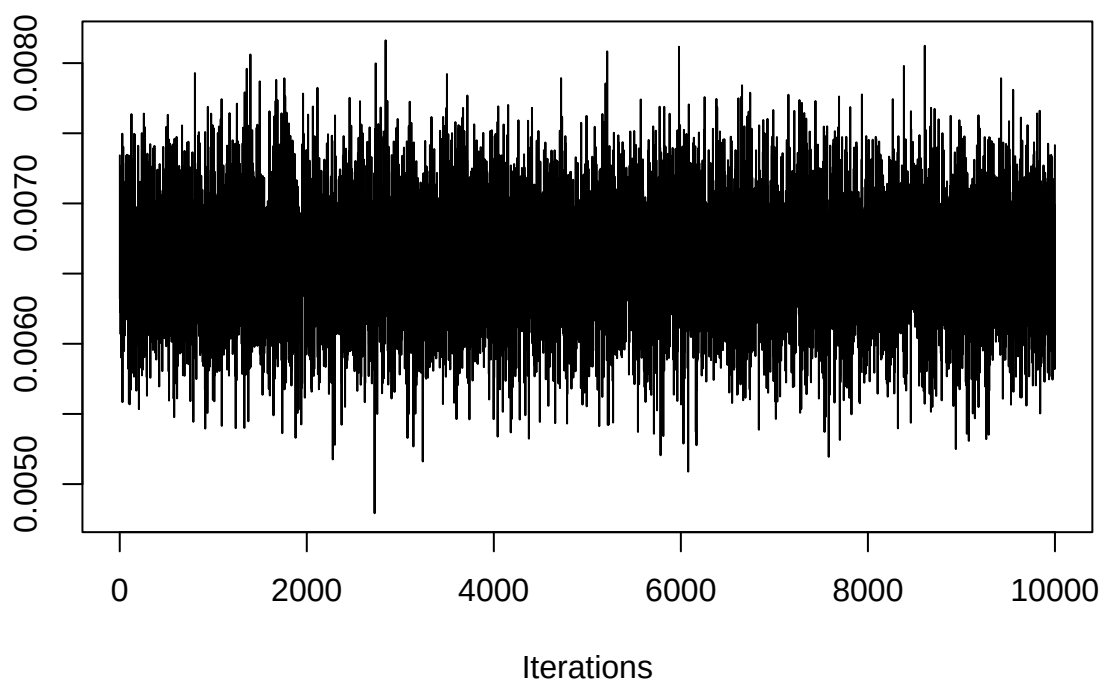
Trace of exclamation_count-exclamation_count



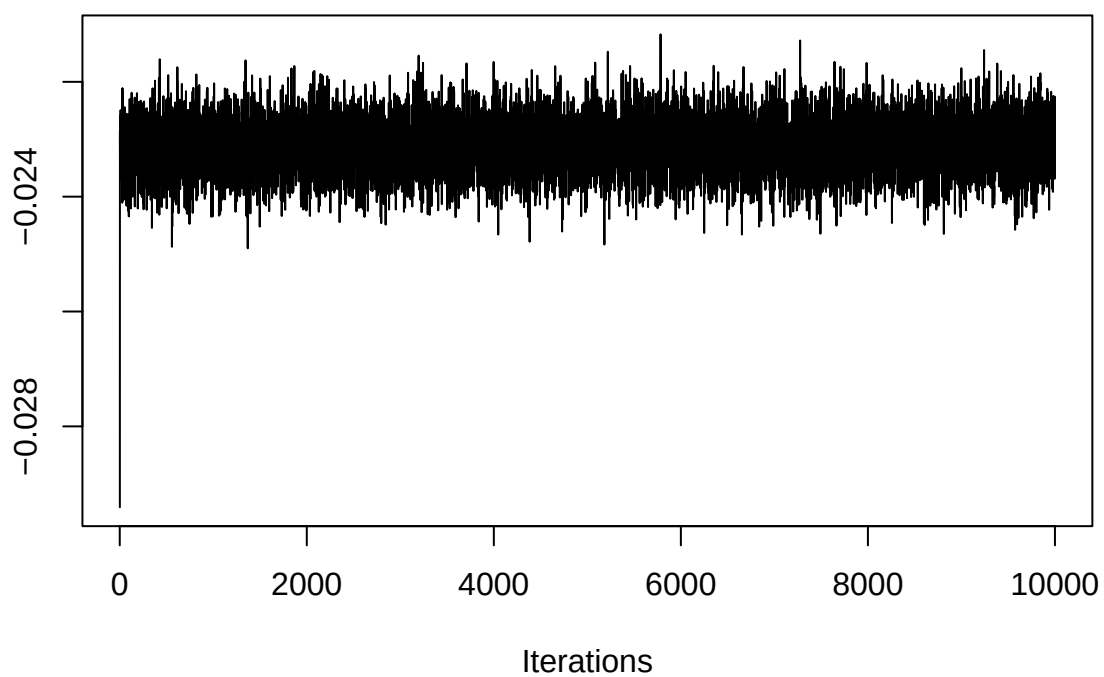
Trace of question_count-question_count



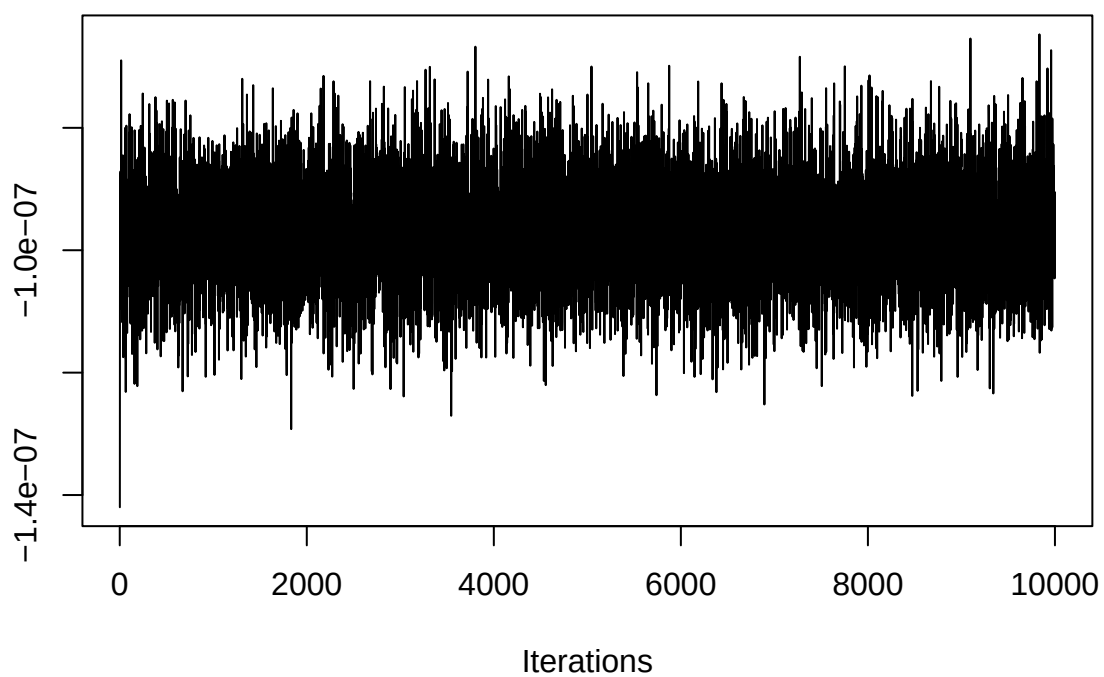
Trace of log_followers-log_followers



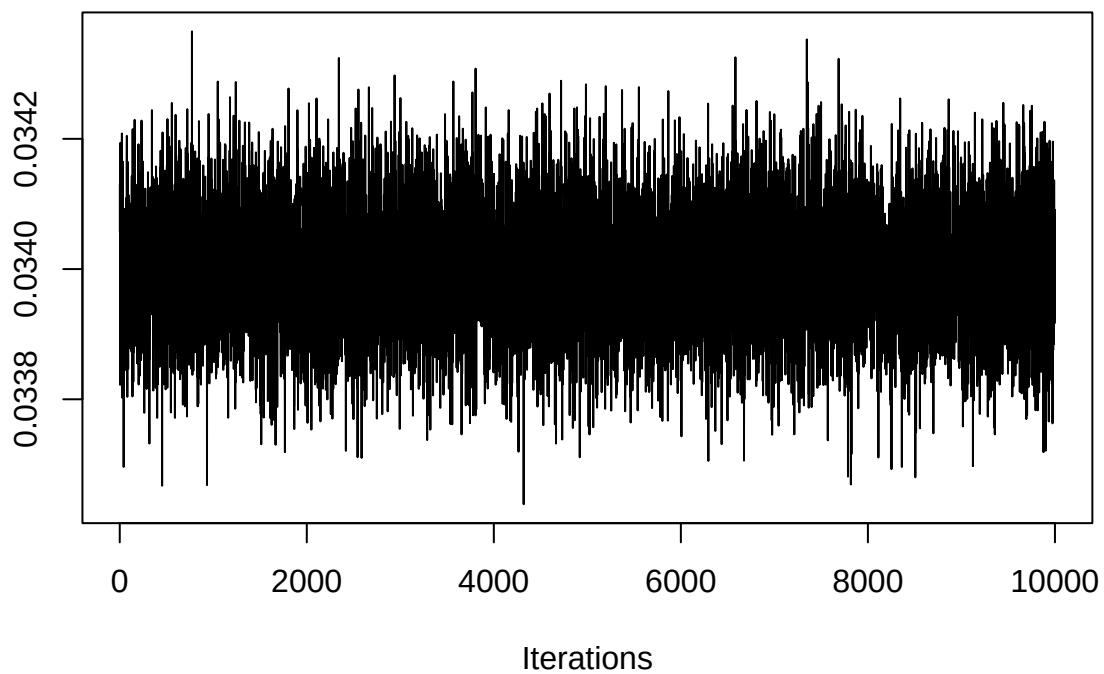
Trace of $\log_friends - \log_friends$

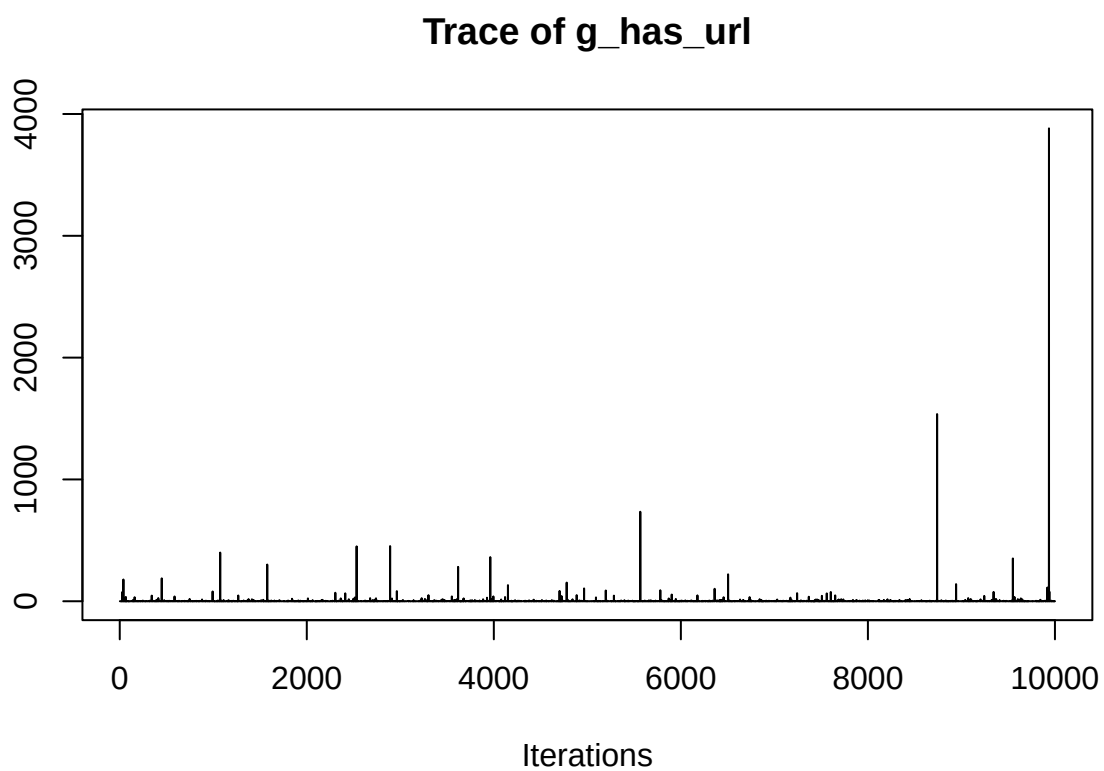


Trace of follower_friend_ratio-follower_friend_ratio

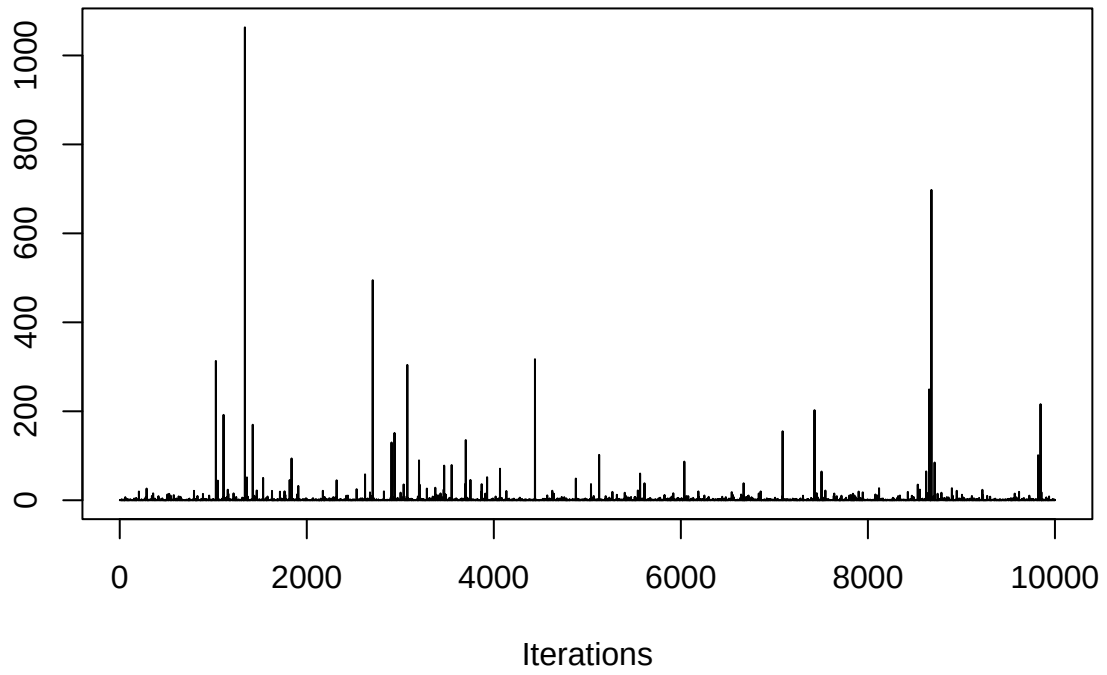


Trace of sig2

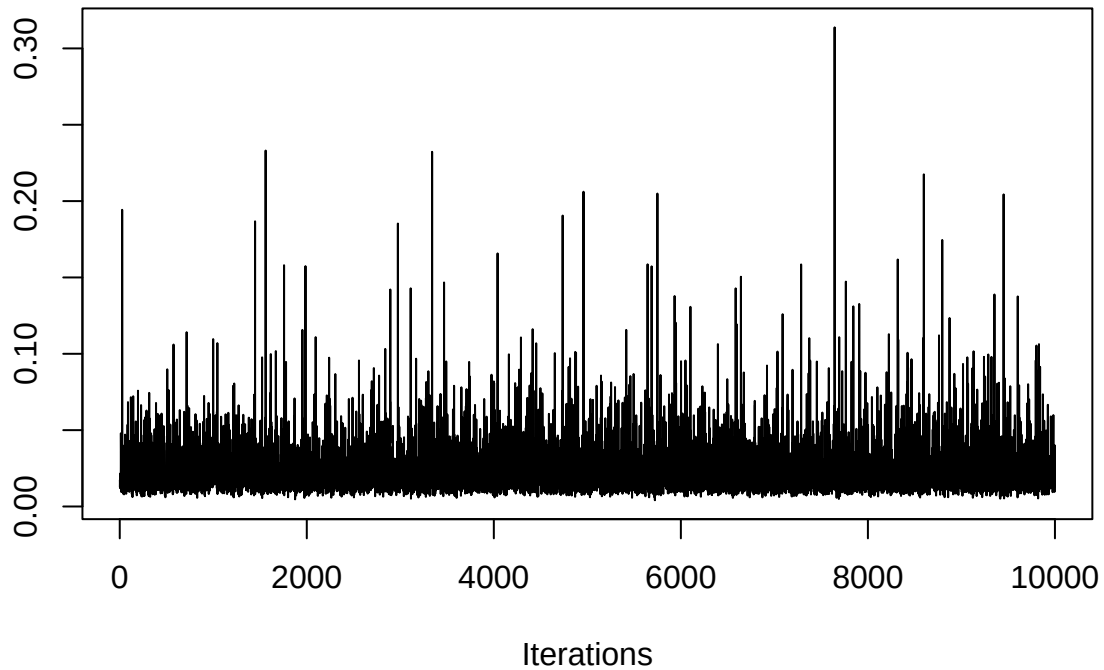




Trace of g_has_mention



Trace of g_continuous



```
describe_posterior(lmout_mcmc)
```

```
## Summary of Posterior Distribution
```

```
##
```

```
## Parameter | Median | 95% CI
```

```
## -----
```

```
## mu | 0.05 | [ 0.05, 0.05]
```

```
## word_count-word_count | -5.44e-05 | [ 0.00, 0.00]
```

```
## hashtag_count-hashtag_count | -5.24e-03 | [-0.01, 0.00]
```

```
## has_url-FALSE | -3.38e-03 | [ 0.00, 0.00]
```

```
## has_url-TRUE | 3.38e-03 | [ 0.00, 0.00]
```

```
## has_mention-FALSE | -4.56e-03 | [-0.01, 0.00]
```

```
## has_mention-TRUE | 4.56e-03 | [ 0.00, 0.01]
```

```
## exclamation_count-exclamation_count | 0.01 | [ 0.01, 0.01]
```

```
## question_count-question_count | -0.02 | [-0.02, -0.02]
```

```
## log_followers-log_followers | 6.59e-03 | [ 0.01, 0.01]
```

```
## log_friends-log_friends | -0.02 | [-0.02, -0.02]
```

```
## follower_friend_ratio-follower_friend_ratio | -9.68e-08 | [ 0.00, 0.00]
```

```
## sig2 | 0.03 | [ 0.03, 0.03]
```

```
## g_has_url | 0.18 | [ 0.03, 5.09]
```

```
## g_has_mention | 0.18 | [ 0.03, 5.02]
```

```
## g_continuous | 0.02 | [ 0.01, 0.07]
```

```
##
```

```
## Parameter | pd | ROPE | % in ROPE
```

```
## -----
```

## mu		100%		[-0.10, 0.10]		100%
## word_count-word_count		94.02%		[-0.10, 0.10]		100%
## hashtag_count-hashtag_count		100%		[-0.10, 0.10]		100%
## has_url-FALSE		100%		[-0.10, 0.10]		100%
## has_url-TRUE		100%		[-0.10, 0.10]		100%
## has_mention-FALSE		100%		[-0.10, 0.10]		100%
## has_mention-TRUE		100%		[-0.10, 0.10]		100%
## exclamation_count-exclamation_count		100%		[-0.10, 0.10]		100%
## question_count-question_count		100%		[-0.10, 0.10]		100%
## log_followers-log_followers		100%		[-0.10, 0.10]		100%
## log_friends-log_friends		100%		[-0.10, 0.10]		100%
## follower_friend_ratio-follower_friend_ratio		100%		[-0.10, 0.10]		100%
## sig2		100%		[-0.10, 0.10]		100%
## g_has_url		100%		[-0.10, 0.10]		27.16%
## g_has_mention		100%		[-0.10, 0.10]		26.95%
## g_continuous		100%		[-0.10, 0.10]		100%

The trace Plots looks good.

The Bayesian model indicates that most predictors have small but meaningful effects on sentiment_compound. Positive associations are observed for exclamation_count, log_followers, and has_mention-TRUE, while negative associations are seen for word_count, hashtag_count, question_count, and log_friends.

For some predictors (word_count and follower_friend_ratio), the 95% CI are very narrow include 0 or super close to 0, suggesting that their effects are weak or potentially negligible. Overall, the model captures subtle but consistent relationships, though practical effect sizes are small.

All posterior distributions are narrow with high probability density, indicating stable estimates, and nearly all parameters lie outside the ROPE of [-0.1, 0.1], confirming meaningful effects. Overall, the model captures subtle but consistent relationships between tweet/user features and sentiment, though effect sizes are small.

Conclusion

Null hyp : No relationship between any predictor and sentiment Alt hyp: At least one predictor significantly affects sentiment

Overall Model Significance: Both analyses show the model is statistically significant (rejecting null hypothesis), indicating at least one predictor affects sentiment. Frequentist R2 is approx 0.018 shows low explanatory power.

Positive Predictors: exclamation_count and log_followers are consistently associated with more positive sentiment.

Negative Predictors: word_count, hashtag_count, and log_friends show negative associations with sentiment, though some intervals include 0 in Bayesian analysis, indicating uncertainty.

Non-Significant: Features like text_length and follower_friend_ratio have minimal or uncertain effects. CIs contain 0.